

MONTE CARLO ASSIGNMENT 3

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1. COAL MINE DISASTERS

In this problem we are going to examine data concerning coal mine disasters in the Great Britain between 1658 and 1980. Let $t_1 = 1658$ and $t_{d+1} = 1980$ be the fixed endpoints of the set and let $t_i, i = 2, \dots, d$ be the breakpoints. Further, let $\mathbf{t} = (t_1, \dots, t_{d+1})$ and let $\lambda = (\lambda_1, \dots, \lambda_d)$, where λ_i be the disaster intensity in the interval $[t_i, t_{i+1})$. The time points of the $n = 751$ disasters are denoted by $\tau = \tau_1, \dots, \tau_n$ and the data between time points t_1 and t_{d+1} is modelled by an inhomogeneous Poisson process with intensity

$$\lambda(t) = \sum_{i=1}^d \lambda_i \mathbb{1}_{[t_i, t_{i+1})}(t),$$

such that the times between each accident are random variables with exponential distributions. Specifically, the interval $[t_i, t_{i+1})$ has the pertaining intensity λ_i . Denote by $n_i(\tau)$ the number of disasters in the sub-interval $[t_i, t_{i+1})$, such that

$$n_i(\tau) = \sum_{j=1}^n \mathbb{1}_{[t_i, t_{i+1})}(\tau_j).$$

1.1. In the first task, the marginal posteriors $f(\theta|\lambda, \mathbf{t}, \tau)$, $f(\lambda|\theta, \mathbf{t}, \tau)$ and $f(\mathbf{t}|\theta, \lambda, \tau)$ are to be computed (up to normalizing constants), and then the related distributions are to be identified.

We know that $\lambda \sim \Gamma(2, 0)$ and $\theta \sim \Gamma(2, \Psi)$, where Ψ is a fixed hyperparameter that needs to be specified. We also know that

$$f(\mathbf{t}) \propto \begin{cases} \prod_{i=1}^d (t_{i+1} - t_i), & \text{for } t_1 < t_2 < \dots < t_d < t_{d+1}, \\ 0, & \text{otherwise.} \end{cases}$$

It follows that

$$f(\tau|\lambda, \mathbf{t}) = \exp\left(-\sum_{i=1}^d \lambda_i (t_{i+1} - t_i)\right) \prod_{i=1}^d \lambda_i^{n_i(\tau)}.$$

We now begin rewriting

$$f(\theta|\lambda, \mathbf{t}, \tau) = \frac{f(\theta, \lambda, \mathbf{t}, \tau)}{f(\lambda, \mathbf{t}, \tau)}$$

by first rewriting the numerator. Applying Bayes' formula several times, we see that

$$f(\theta, \lambda, \mathbf{t}, \tau) = f(\tau|\theta, \lambda, \mathbf{t})f(\theta, \lambda, \mathbf{t}) = f(\tau|\theta, \lambda, \mathbf{t})f(\mathbf{t}|\theta, \lambda)f(\theta, \lambda) = f(\tau|\theta, \lambda, \mathbf{t})f(\mathbf{t}|\theta, \lambda)f(\lambda|\theta)f(\theta).$$

We now note that τ is independent of θ , and that $\mathbf{t}$ is independent of both λ and θ . Thus

$$f(\theta, \lambda, \mathbf{t}, \tau) = f(\tau|\lambda, \mathbf{t})f(\mathbf{t})f(\lambda|\theta)f(\theta).$$

Now we use proportionality, and the fact that we are only looking for the marginal posteriors up to a normalizing constant, to write

$$f(\theta|\lambda, \mathbf{t}, \tau) = \frac{f(\tau|\lambda, \mathbf{t})f(\mathbf{t})f(\lambda|\theta)f(\theta)}{f(\lambda, \mathbf{t}, \tau)} \propto f(\lambda|\theta)f(\theta),$$

where the terms not dependent on θ has been removed. Finally, we rewrite $f(\lambda|\theta)f(\theta)$ using our knowledge of $f(\lambda|\theta)$ and $f(\theta)$;

$$f(\theta|\lambda, \mathbf{t}, \tau) \propto f(\lambda|\theta)f(\theta) = \left(\prod_{i=1}^d \frac{\theta^2}{\Gamma(2)} \lambda_i e^{-\theta \lambda_i} \right) \frac{\Psi^2}{\Gamma(2)} \theta e^{-\Psi \theta} = \left(\prod_{i=1}^d \lambda_i \right) \frac{\Psi^2}{\Gamma(2)} \theta^{2d+1} e^{-\theta(\Psi + \sum_{i=1}^d \lambda_i)},$$

which we recognize as the density of a Gamma distribution. Specifically,

$$f(\theta|\lambda, \mathbf{t}, \tau) \sim \Gamma\left(2d + 2, \Psi + \sum_{i=1}^d \lambda_i\right).$$

For the second marginal posterior, $f(\lambda|\theta, \mathbf{t}, \tau)$, we again use proportionality to see that

$$f(\lambda|\theta, \mathbf{t}, \tau) \propto f(\tau|\lambda, \mathbf{t})f(\lambda|\theta) = \prod_{i=1}^d e^{-\lambda_i(t_{i+1} - t_i + \theta)} \lambda_i^{n_i(\tau) + 1}$$

such that

$$f(\lambda|\theta, \mathbf{t}, \tau) \sim \Gamma(n_i(\tau) + 2, t_{i+1} - t_i + \theta).$$

For the last marginal posterior $f(\mathbf{t}|\theta, \lambda, \tau)$, we have

$$f(\mathbf{t}|\theta, \lambda, \tau) \propto f(\tau|\lambda, \mathbf{t})f(\mathbf{t}) = \prod_{i=1}^d (t_{i+1} - t_i) e^{-\lambda_i(t_{i+1} - t_i)} \lambda_i^{n_i(\tau) + 1},$$

for which we are unable to recognize any specific distribution.

1.2. In this section, a hybrid MCMC algorithm that samples from the posterior $f(\theta, \lambda, \mathbf{t}|\tau)$ will be constructed. The Gibbs sampler will be used to update the posteriors pertaining to θ and λ , since their distributions are known. In combination, the Metropolis-Hastings (MH) algorithm will be used to update the breakpoints $\mathbf{t}$, since we don't have access to the posterior distribution.

To use the Gibbs sampler to update the marginal posteriors $f(\theta|\lambda, \mathbf{t}, \tau)$ and $f(\lambda|\theta, \mathbf{t}, \tau)$, for which the distributions are known, suppose that the state space can be divided into m blocks. Then, given X_k , a Markov chain can be formed according to the

following mechanism

$$\begin{aligned}
&\rightarrow X_{k+1}^1 \sim f_1(x^1 | X_k^2, \dots, X_k^m) \\
&\rightarrow X_{k+1}^2 \sim f_2(x^2 | X_{k+1}^1, X_k^3, \dots, X_k^m) \\
&\rightarrow X_{k+1}^3 \sim f_3(x^3 | X_{k+1}^1, X_{k+1}^2, X_k^4, \dots, X_k^m) \\
&\rightarrow \dots \\
&\rightarrow X_{k+1}^m \sim f_m(x^m | X_{k+1}^1, X_{k+1}^2, \dots, X_{k+1}^{m-1}).
\end{aligned}$$

For the marginal posterior $f(\mathbf{t}|\theta, \lambda, \tau)$, for which we don't know the distribution, the MH algorithm will be utilized. The MH algorithm works as follows: given an X_k ,

$$\begin{aligned}
&\rightarrow \text{generate } X^* \sim r(z | X_k) \text{ and} \\
&\rightarrow \text{set } X_{k+1} = \begin{cases} X^* & \text{w. pr. } \alpha(X_k, X^*) \stackrel{\text{def.}}{=} 1 \wedge \frac{f(X^*)r(X_k | X^*)}{f(X_k)r(X^* | X_k)}, \\ X_k & \text{otherwise.} \end{cases}
\end{aligned}$$

where $a \wedge b \stackrel{\text{def.}}{=} \min\{a, b\}$. The proposal is chosen as the symmetric *Random walk proposal one at a time*. This means that we update one breakpoint at a time in the following way: for each breakpoint t_i , we generate a candidate t_i^* according to

$$t_i^* = t_i + \varepsilon, \quad \text{where } \varepsilon \sim \mathcal{U}(-R, R),$$

and

$$R = \rho(t_{i+1} - t_{i-1}),$$

where ρ is a tuning parameter that determines the scale of the random walk step for updating single breakpoint t_i . Clearly, $t_{i-1} < t_i^* < t_{i+1}$, since t_{i-1} and t_{i+1} are fixed each time. This make sure the chain remains in the valid space of ordered breakpoints.

The proposal is a valid symmetric proposal for the MH algorithm since ϵ is uniformly drawn from the interval $(-R, R)$ such that for any allowed t_i^* ,

$$q(t_i^* | t_i) = \frac{1}{2R} = q(t_i | t_i^*).$$

This further implies that the MH acceptance probability $\alpha(X_k, X^*)$ is reduced to

$$\alpha(X_k, X^*) = 1 \wedge \frac{f(X^*)}{f(X_k)}.$$

1.3. We now have attained the posterior distributions and the hybrid MCMC method (using Gibbs sampling for updating θ and λ , using MH algorithm to update t) from 1.1 and 1.2. Our task now is to estimate the chain for the number of break points. Notice that for $d-1$ breakpoints to be estimated, there are d homogeneous distributions with different parameters to be estimated. We first load the data and observe Figure 1. A direct observation is that d is at least 4, since trend changes through from 1658 to around 1760 the disasters is almost 0 and the increasing rate is slow followed by from 1760 to around 1860 the increasing rate is getting faster and faster and from around 1860 to the end the increasing rate started to decrease.

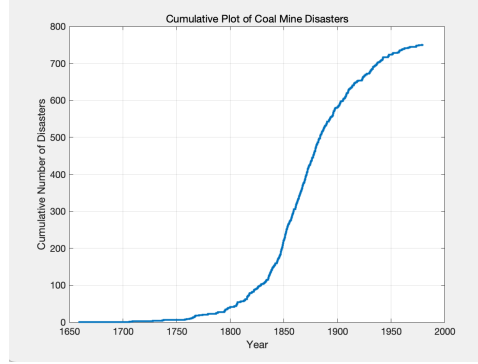
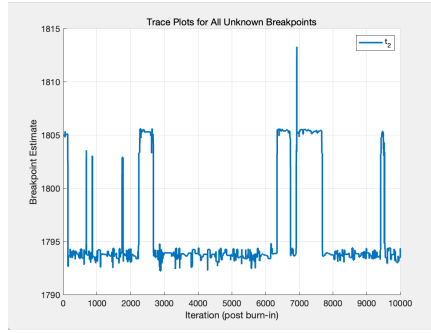
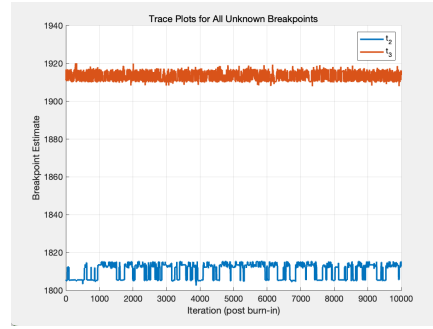
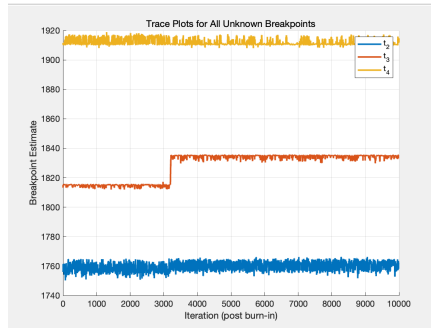
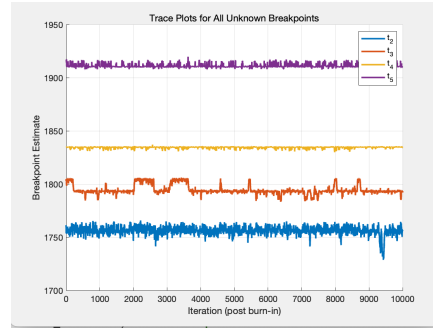


FIGURE 1. Total number of disasters vs year

When $d = 1$, there is nothing to estimate. Therefore, we start by investigating when $d = 2, 3, 4, 5$ with 12000 samples and discard the first 2000 samples, this is because the algorithm takes some time to update from initial 'guess' of the parameter $\Psi = 10$. Below shows the results of our estimation with different numbers of breakpoints to be estimated:

(A) $d=2$, 1 breakpoints(B) $d=3$, 2 breakpoints(C) $d=4$, 3 breakpoints(D) $d=5$, 4 breakpoints(E) $d=6$, 5 breakpointsFIGURE 2. Comparison of breakpoints for different values of d .

Base on estimations, we think $d = 6$ with 5 breakpoints is reasonable and more stable as well as it matches with our observation in the beginning. The estimated breakpoints from hybrid MCMC method gives:

Breakpoint
1658
1756
1796
1835
1891
1943
1980

TABLE 1. Posterior means of breakpoints.

1.4. We now investigate how sensitive are the posteriors to the choice of hyperparameter Ψ . Recall that the prior of θ is $\theta \sim \Gamma(2, \Psi)$. and then since each λ_i has prior $\Gamma(2, \theta)$, we get a likelihood contribution $\theta^2 e^{-\theta \lambda_i} \lambda_i^1$. Combined over d intervals, plus the hyperprior $\Gamma(2, \Psi)$, yields:

$$\theta \mid (\lambda_1, \dots, \lambda_d, \tau) \sim \Gamma\left(2 + 2d, \Psi + \sum_{i=1}^d \lambda_i\right).$$

Hence the posterior mean and variance are:

$$\mathbb{E}[\theta \mid \lambda] = \frac{2 + 2d}{\Psi + \sum_i \lambda_i}, \quad \text{Var}(\theta \mid \lambda) = \frac{2 + 2d}{(\Psi + \sum_i \lambda_i)^2}.$$

As we see in the expression above, larger choices for Ψ will yield smaller $\mathbb{E}[\theta \mid \lambda]$ and lower variance because Ψ is in the denominator.

The prior of λ_i is $\lambda_i \sim \Gamma(2, \theta)$, and the likelihood is Poisson-like with $\lambda_i^{n_i(\tau)} e^{-\lambda_i \Delta_i}$, where Δ_i is the interval length $\Delta_i = (t_{i+1} - t_i)$. Combining these we have:

$$\lambda_i \mid (\theta, t, \tau) \sim \Gamma(n_i(\tau) + 2, \Delta_i + \theta).$$

Hence the posterior mean and variance (conditional on θ, t, τ) are:

$$\mathbb{E}[\lambda_i \mid \theta, t, \tau] = \frac{n_i(\tau) + 2}{\Delta_i + \theta}, \quad \text{Var}(\lambda_i \mid \theta, t, \tau) = \frac{n_i(\tau) + 2}{(\Delta_i + \theta)^2}.$$

As we see in the expression above, since λ_i depends of θ , so consequently it depends on the choice of Ψ . Larger Ψ will give us smaller θ and hence larger λ_i estimates with slightly larger variances since Δ_i is the dominating factor.

For t , the prior is given by $f(t) \propto \prod_{i=1}^d (t_{i+1} - t_i)$, for $t_1 < t_2 < \dots < t_{d+1}$ with likelyhood function $\lambda_i^{n_i(\tau)} \exp\{-\lambda_i \Delta_i\}$. So the posterior for t (up to a constant) is:

$$\pi(t \mid \lambda, \theta, \tau) \propto f(\tau \mid \lambda, t) f(t).$$

Substituting the expressions we get:

$$f(t \mid \lambda, \theta, \tau) \propto \prod_{i=1}^d \left[\lambda_i^{n_i(\tau)} \exp\{-\lambda_i \Delta_i\} (\Delta_i) \right].$$

As we see in the expression above, the posterior will be affected by Ψ because λ_i are affected by Ψ . Larger Ψ will give us larger λ_i . However once again since Δ_i is the dominating factor, the posterior is not likely to be affected by the choice of Ψ .

1.5. We now investigate how the choice of ρ affects the mixing and posteriors to the final estimation of t_i . ρ can be understood as the step size, apparently smaller ρ makes the acceptance rate higher because small moves are less likely to drastically reduce the posterior density. The chain will move slower through the space of breakpoints. On the other hand, if ρ is big, meaning that the step size is larger the acceptance rate will be smaller the chain might 'bounce around' with frequent rejections and not converge smoothly.

However, regardless of the choice of ρ , the chain will still converge in the end because the chain is stationary as analyzed in 1b. Therefore, the posterior is only neglectfully affected by the choice of ρ .

A moderate acceptance rate of the chain is often around 30, after trying ρ with values from 0.01 to 0.1, our final choice is $\rho = 0.034$ with acceptance rate 32.71%.

2. PARAMETRIC BOOTSTRAP FOR THE 100-YEAR ATLANTIC WAVE

Data of significant wave-height have been recorded 14 times a month for several winter months in the north Atlantic. The expected 100-year return value of the significant wave-height gives the largest expected value during a 100-year period. The *Gumbel distribution* is a good fit to the data. The T th return value is given by $F^{-1}(1 - 1/T; \mu, \beta)$, where $T = 3 \times 14 \times 100 = 4200$.

2.1. Given the Gumbel distribution function

$$F(x; \mu, \beta) = \exp\left(-\exp\left(-\frac{x - \mu}{\beta}\right)\right), \quad \mu, x \in \mathbb{R}, \beta > 0,$$

we are going to find the inverse function $F^{-1}(u; \mu, \beta)$.

To this end, let $u = F(x; \mu, \beta)$ such that $x = F^{-1}(u; \mu, \beta)$. Then

$$\begin{aligned} u &= \exp\left(-\exp\left(-\frac{x - \mu}{\beta}\right)\right) \\ \implies \ln(-\ln(u)) &= -\frac{x - \mu}{\beta} \\ \implies x &= \mu - \beta \ln(-\ln(u)) \end{aligned}$$

such that

$$F^{-1}(u; \mu, \beta) = \mu - \beta \ln(-\ln(u))$$

is the inverse of $F(x; \mu, \beta)$.

2.2. We are now going to construct 99% confidence intervals for the parameters μ and β using parametric bootstrap. This is done by first acquiring estimates $\hat{\mu}$ and $\hat{\beta}$ of the parameters using MATLAB's inbuilt function *est_gumbel.m*. The acquired estimates are given by

$$\begin{aligned} \hat{\mu} &= 4.1477, \\ \hat{\beta} &= 1.4858. \end{aligned}$$

Together with these estimates, we use $N = 1000$ bootstrap resamples when creating the confidence intervals. To resample, we generate random uniform samples u from the interval $(0, 1)$ and then transform them using the Gumbel inverse $F^{-1}(u; \mu, \beta)$.

Then, using the same method as on page 33 in Lecture notes 13, we create 99% confidence intervals for the parameters. In other words, we sort the differences between the first and the bootstrapped estimates to obtain the desired quantiles. The following table contains the acquired 99% confidence intervals.

Parameter	Interval
μ	(3.988, 4.3202)
β	(1.3593, 1.6151)

TABLE 2. 99% confidence intervals for parameters μ and β

2.3. In the final task, we want to create a one-sided upper bounded parametrically bootstrapped 99% confidence interval for the 100-year return value. To do this, we use our knowledge that the T th return value is given by $F^{-1}(1 - 1/T; \mu, \beta)$, where $T = 3 \times 14 \times 100 = 4200$. First, an estimate of the wave-height is acquired with the help of $\hat{\mu}$ and $\hat{\beta}$ from the previous section (i.e., $F^{-1}(1 - 1/T; \hat{\mu}, \hat{\beta})$). Then we make a new estimation using the previously bootstrapped parameters. The 99% upper bound is then acquired in the same way the upper bound was acquired for the previous confidence intervals; sorting the differences to obtain quantiles to then get the upper bound. In the following table the estimate is given together with the obtained 99% upper bound.

Estimate	99% Upper bound
16.5436	17.5581

TABLE 3. 99% upper bound for the 100-year return value